KOP Homework 02: Maximum Weighted Satisfiability via Simulated Annealing

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Executive Summary

Simulated annealing with adaptive initial temperature, equilibrium length, and derived cooling rate achieves >97% satisfiability and >89% optimality on MWSAT instances with 20–50 variables. The adaptive mechanisms enable consistent performance across instance sizes without manual parameter tuning.

1 White-Box Phase: Algorithm Development and Parameter Tuning

This section documents the iterative development process and the rationale behind parameter choices.

1.1 Algorithm Overview

The algorithm uses simulated annealing to solve Maximum Weighted SAT (MWSAT). The energy function combines satisfiability constraints with the weighted objective. A natural formulation would be:

$$E' = M \cdot (\text{unsatisfied clauses}) - (\text{weighted objective})$$

where $M = 1 + \sum_i w_i$ ensures feasibility dominates until all clauses are satisfied. However, since M can be large (sum of all weights), this leads to energy values with high magnitude, making temperature scaling difficult. Instead, we divide by M:

$$E = \underbrace{(\text{unsatisfied clauses})}_{\text{feasibility}} - \underbrace{\frac{\text{weighted objective}}{M}}_{\text{optimality}}$$

This preserves the property that any satisfying assignment has lower energy than any non-satisfying one, while keeping energy differences in a reasonable range for the Metropolis criterion.

1.2 Initial Configuration

Parameter	Initial Value
Initial temperature T_0	Computed adaptively (see §1.4)
Cooling schedule	Geometric: $T_{k+1} = \alpha \cdot T_k$
Cooling rate α	0.95
Equilibrium length L_{eq}	$2n$
Stopping criterion	$T < T_{min}$

1.3 Parameter Tuning Experiments

All tuning experiments were performed on an Apple M4 Pro (8 performance cores).

1.3.1 Baseline ($L_{eq} = 2n$, $\alpha = 0.95$)

Dataset	Solved (%)	Optimal (%)
wuf20-91-M	92.63	82.18
wuf50-218-M	63.72	43.38

Performance degrades on larger instances due to insufficient exploration.

1.3.2 Increased Equilibrium Length ($L_{eq} = 20n$)

Dataset	Solved (%)	Optimal (%)
wuf20-91-M	95.04	88.76
wuf50-218-M	72.24	51.46

Improvement observed, but still insufficient for larger instances.

1.3.3 Adaptive Equilibrium Length

Fixed L_{eq} either wastes time on small instances or underperforms on large ones. The equilibrium length now adapts based on acceptance rate:

$$L_{eq} = \text{clamp}\left(\frac{c \cdot n}{a}, n, 50n\right)$$

where a is the acceptance rate from the previous temperature level and c is a tuning constant.

Dataset	Solved (%)	Optimal (%)
wuf20-91-M	100.00	99.91
wuf50-218-M	92.13	77.20

Significant improvement. Adaptive mechanism responds to instance difficulty.

1.3.4 Cooling Rate ($\alpha = 0.99$)

Slower cooling allows finer exploration near optimal regions.

Dataset	Solved (%)	Optimal (%)
wuf50-218R-M	99.16	96.97
wuf50-218R-Q	93.37	74.57

Substantial improvement on difficult instances.

1.3.5 Number of Temperature Levels (K)

Instead of fixing α, we fix K (number of temperature levels) and derive α from it.

Dataset	K=300 Opt (%)	K=700 Opt (%)	K=900 Opt (%)
wuf50-218R-M	95.65	99.69	99.88
wuf50-218R-Q	69.33	90.59	93.98

K=900 provides best results with acceptable runtime.

1.3.6 Tuning Constant c for Adaptive L_{eq}

Dataset	$c = 12$ Opt (%)	$c = 15$ Opt (%)
wuf50-218R-M	99.88	99.90
wuf50-218R-Q	92.50	93.99

Final choice: $c = 12$ (comparable performance, faster runtime).

1.4 Adaptive Mechanisms

1.4.1 Initial Temperature Computation

The initial temperature T_0 is computed to achieve a target acceptance probability $P_0 = 0.99$ for worsening moves. The algorithm samples 1000 random moves, collects energy increases from worsening moves, then iteratively solves for T such that the mean acceptance probability for these worsening moves equals P_0.

1.4.2 Adaptive Equilibrium Length

As described in §1.3.3, the equilibrium length adapts based on acceptance rate.

1.4.3 Derived Cooling Rate

The cooling rate α is computed to achieve exactly K temperature levels:

$$\alpha = \left(\frac{T_{min}}{T_{max}}\right)^{1/K}$$

1.5 Convergence Behavior

Figure 1 shows a single run in detail, illustrating the characteristic SA behavior: high variance early (exploration at high temperature) gradually reducing as temperature decreases (exploitation). The algorithm first reduces unsatisfied clauses toward zero, then optimizes the weighted objective.

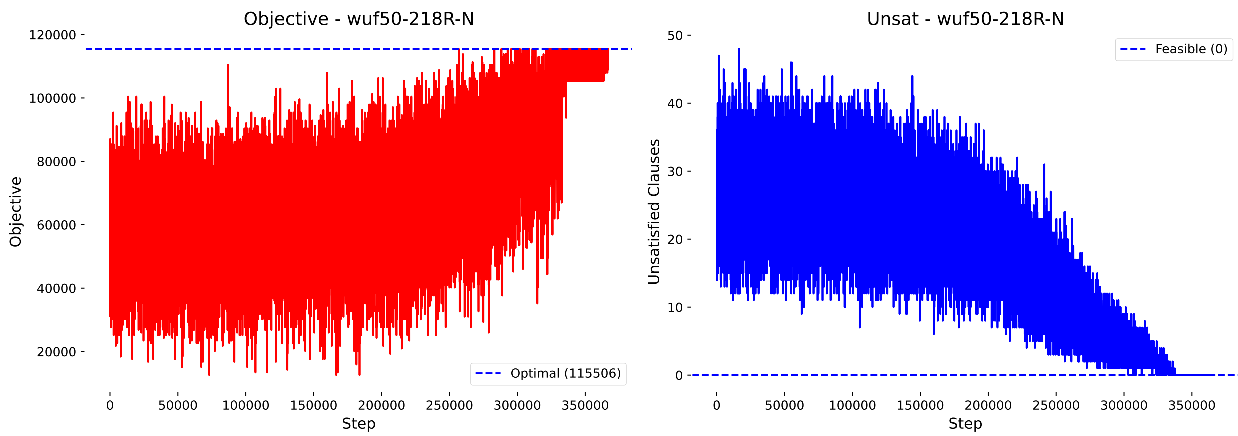

Figure 1: Single run convergence on wuf50-218R-N showing objective (left) and unsatisfied clauses (right). The decreasing variance reflects the cooling schedule.

Figure 2 shows 10 independent runs on the same instance, each starting from a different random initial assignment. Despite different starting points, all runs converge to the same region and reach feasibility. The bottom row zooms in on the first 5000 steps, showing that after initial divergence the runs settle into similar trajectories. This consistency across random starts suggests the algorithm has iterative strength—it reliably finds good solutions regardless of initialization. Note that Figure 2 uses a moving average for both objective and unsatisfied clauses, which is why the curves appear short of the optimal value; the raw values (shown in Figure 1) are noisy but do reach the optimum.

Note that the algorithm uses zero restarts. In practice, running 2–5 independent restarts and taking the best result would further increase the probability of finding the optimal assignment.

1.6 Final Algorithm Configuration

Parameter	Value
Initial temperature	Computed adaptively ($P_0 = 0.99$)
T_{min}/T_{max} ratio	10^{-4}
Number of temperature levels K	900
Cooling rate α	Derived from K
Equilibrium length L_{eq}	$\mathrm{clamp}(12n/a, n, 50n)$
Target accepted moves	$\min(L_{eq}, 12n)$

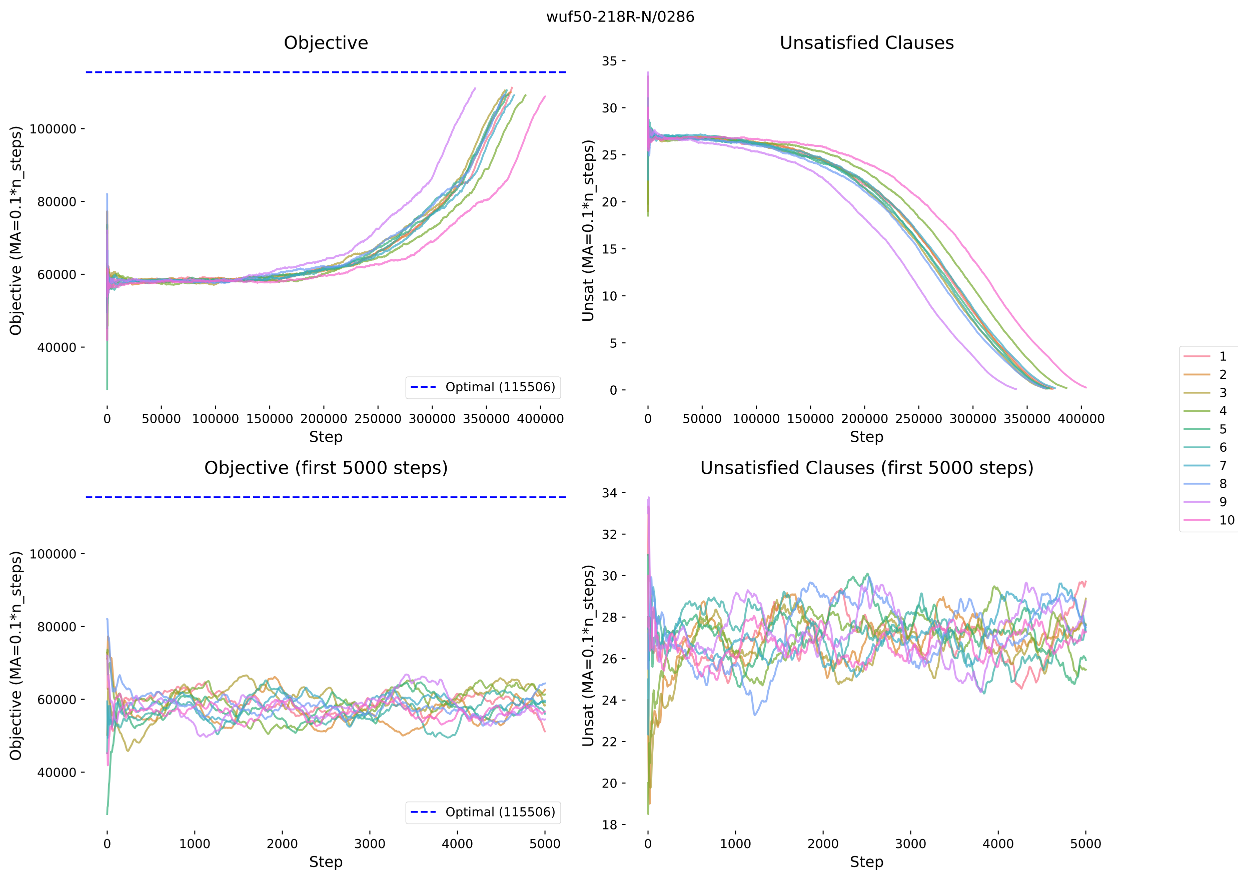

Figure 2: Convergence of 10 independent runs on wuf50-218R-N/0286. Top: full run showing all runs reaching feasibility and converging toward optimal. Bottom: zoom on first 5000 steps showing runs settling into similar trajectories despite random starts.

2 Black-Box Phase: Experimental Evaluation (IMRAD)

2.1 Introduction

We evaluate the final simulated annealing configuration on the MWSAT problem to determine its effectiveness across instance sizes and characteristics. The heuristic should achieve comparable performance on instances with 20–50 variables without manual parameter adjustment.

Research questions:

1. What is the success rate and optimality rate across instance classes?
2. Does the adaptive mechanism provide consistent performance across instance sizes?

2.2 Methods

2.2.1 Algorithm Summary

The simulated annealing algorithm uses a weighted energy function prioritizing feasibility over optimality, adaptive initial temperature based on acceptance probability, geometric cooling with 900 temperature levels, adaptive equilibrium length based on acceptance rate, and early termination when a valid solution is found and acceptance rate drops to zero.

2.2.2 Instance Sets

Instance sets from the course page: wruf36-157, wuf20-71R, wuf20-91, wuf20-91R, wuf50-218, and wuf50-218R. These sets were chosen because they provide reference optimal assignments, allowing direct comparison of solver results against known optima.

2.2.3 Experimental Protocol

To control for randomization, experiments follow a two-stage protocol. First, suppress randomization variance by running the solver $r = 100$ times on each instance. Second, suppress instance variance by aggregating over all instances in each set. Total: $13200 \times 100 = 1.32 \times 10^6$ solver invocations.

2.2.4 Metrics

- **Solved (%)**—percentage of runs finding a satisfying assignment.
- **Optimal (%)**—percentage of runs finding the known optimal solution.

2.2.5 Implementation

The solver is implemented in C++17. Batch execution uses GNU Parallel [1] to run one binary per instance file, where each binary performs all r repetitions internally. This reduces parallelization overhead since individual SA runs complete quickly and spawning a new process per run would leave CPUs idle waiting for jobs.

Final evaluation was performed on Metacentrum using a 32-core AMD EPYC 7543 (roughly 45 minutes).

Table 1: Final evaluation results across all instance sets.

Dataset	Solved (%)	Optimal (%)
wuf20-71R-M	100.00	100.00
wuf20-71R-N	100.00	100.00
wuf20-71R-Q	100.00	99.55
wuf20-71R-R	100.00	99.55
wuf20-91-M	100.00	100.00
wuf20-91-N	100.00	100.00
wuf20-91-Q	100.00	100.00
wuf20-91-R	100.00	100.00
wuf20-91R-M	100.00	100.00
wuf20-91R-N	100.00	100.00
wuf20-91R-Q	100.00	100.00
wuf20-91R-R	100.00	100.00
wruf36-157-M	100.00	99.99
wruf36-157-N	100.00	99.99
wruf36-157-Q	99.75	99.04
wruf36-157-R	99.76	99.06
wuf50-218-M	99.94	99.56
wuf50-218-N	99.92	99.55
wuf50-218-Q	98.12	90.63
wuf50-218-R	98.20	90.59
wuf50-218R-M	99.92	99.63
wuf50-218R-N	99.95	99.61
wuf50-218R-Q	97.46	89.76
wuf50-218R-R	97.69	89.93

2.3 Results

2.4 Discussion

The heuristic achieves 100% satisfiability and near-100% optimality on 20-variable instances, >99% satisfiability on 36 and 50-variable instances with M/N weights, and >97% satisfiability on difficult Q/R weight distributions. Optimality rates remain above 89% even on the most challenging instances.

Performance degrades gracefully from 20 to 50 variables, demonstrating that adaptive mechanisms successfully adjust to instance difficulty.

The convergence plots (Figure 1, Figure 2) demonstrate iterative strength: the algorithm consistently improves solutions over time, with all runs reaching feasibility and most reaching optimality.

2.5 Conclusion

The simulated annealing heuristic with adaptive mechanisms achieves >97% satisfiability and >89% optimality across all tested instance classes (20–50 variables). The adaptive initial temperature, equilibrium length, and derived cooling rate enable consistent performance without manual parameter tuning per instance.

Acknowledgments

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References

[1] Ole Tange. *GNU Parallel 20251222 ('Bondi')*. Zenodo, December 2025. `https://doi.org/10.5281/zenodo.18039569`

[2] *KOP08: Simulované ochlazování* [Lecture slides]. NI-KOP, FIT CVUT, 2023.